Q.P. Code : 123221

St Philomena's College (Autonomous), Mysuru

II Semester BCA – Final Examination : May / June – 2025

Subject: COMPUTER APPLICATIONS

Title: Data Structures using C++

Time: 3 Hours **Maximum Marks: 80**

PART – A

Answer any TEN of the following : **10 × 2 = 20**

1. Define Algorithm.
2. List the goals of Data Structures.
3. Define Stack.
4. Define D Queue.
5. Write the Logical representation.
6. List the applications of Queue.
7. State the difference between Dynamic Memory Allocation and Static Memory allocation.
8. Define Circular Linked List. Write the memory representation.
9. Define Complete Binary tree.
10. Define Heap.
11. Define weighted Graph.
12. List the different Sorting techniques

PART-B

Answer any SIX of the following : **6 × 10 = 60**

13. Explain the classification of Data Structures with a neat diagram

14. a. Write a program to illustrate Stack operation. **6**

    b. List the applications of Stacks. **4**

15. Define Arrays. Explain the different operations that can be performed on Arrays.

16. Define Queues. Explain operations that can be performed on Linear Queues with memory representation.

17. a. Explain addition of a node in a Linked List at the beginning with memory representation. **6**

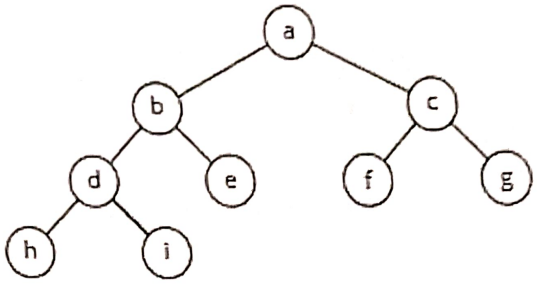
    b. Write note on Garbage Collection. **4**

18. Explain the following tree terminology with an diagram

    a) Height b) Leaf node c) Siblings d) Path e) Degree of a node

PTO

19. List the tree traversal techniques perform the tree traversal for the following tree.



20. a. Explain Adjacency Matrix with an example 5

b. Define the following:

i) Vertex ii) Edge iii) Directed Graph iv) Path v) Cyclic Graph 5

21. a. Define Time and space Complexity. 2

b. Write Binary Search program for searching an element in an array. 8

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Q.P. Code : 123223

St Philomena's College (Autonomous), Mysuru

II Semester BCA - Final Examination : May / June – 2025

Subject: COMPUTER APPLICATIONS

Title: Operating Systems

S240232

Time: 3 Hours

Maximum Marks: 80

PART – A

Answer any TEN Questions 10×2=20

1. Define Operating System
2. What is multiprogramming?
3. What is cooperating process?
4. Define Deadlock.
5. What are the two types of Readers-Writers problem?
6. What is resource allocation graph?
7. What is the difference between logical and physical address space?
8. Define segmentation paging.
9. What is dynamic linking?
10. What is the use of the expr command in shell scripting?
11. What is the purpose of the vi editor?
12. What is the difference between while and until loops in shell Scripting?

PART – B

Answer any SIX Questions 6×10=60

13. Explain in detail the various services provided by an operating system.
14. What is a process? Explain the process states with a process state diagram.
15. Consider the following processes with their burst times:

Process	Burst Time (in ms)
P1	10
P2	5
P3	8

The time quantum is 4 ms.

Calculate Waiting Time and Turn Around Time using Round Robin CPU Scheduling Algorithm

16. What are semaphores? Explain their types and how they are used in process synchronization?

PTO

17. Define Deadlocks. Explain necessary conditions for deadlock Avoidance.

18. Explain the different types of files and list file access.

19. Explain contiguous memory allocation and discuss its types.

20. a. Explain the architecture of the UNIX operating system with a neat diagram. 06

b. Explain any four shell commands. 04

21. Explain looping statements in shell programming with example.

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Q.P. Code : 123222

St Philomena's College (Autonomous), Mysuru

II Semester BCA – Final Examination : May / June – 2025

Subject: COMPUTER APPLICATIONS

Title: Object Oriented Programming with Java

Time: 3 Hours **Maximum Marks: 80**

PART – A

Answer any TEN of the following : $10 \times 2 = 20$

1. What are the basic principles of Object-Oriented Programming?
2. Expand JDK, JVM, and JRE.
3. Write the purpose of the final keyword in Java.
4. Define method overloading.
5. How does this keyword work in Java?
6. Write the syntax of two-dimensional arrays.
7. Mention different states of a thread in Java.
8. Define multithreading.
9. How does the finally block work in exception handling?
10. Write any two differences between Radio Button and Checkbox in Java.
11. Define layout managers in AWT.
12. Define file and stream in Java.

PART-B

Answer any SIX of the following : $6 \times 10 = 60$

13. Explain the basic principles of Object-Oriented Programming.
14. Write a program to demonstrate single intendance.
15. Explain constructors in Java. Provide examples.
16. Explain looping statements in Java.
17. Explain the life cycle of a thread with neat diagram.
18. Discuss exception handling in Java with an example program.
19. Explain the different types of file streams in Java.
20. Explain different controls used in GUI programming.

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